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Project Title:	Human Multi-Tissue Platform to Study Effects of Space Radiation and Countermeasures		
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Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
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Human Research Program Risks:	None		
Space Biology Element:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 TRISH Space Radiation Solicitation TSRAD-2020. Translational Research Institute for Space Health (TRISH) Human-Based Models to Study Effects of Space Radiation and Countermeasures
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Key Personnel Changes/Previous PI:			
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Task Description:	The exact effects of space radiation, a potentially serious risk encountered during prolonged missions to Moon and Mars ("red risk"), are still uncertain. There is a compelling need to better understand the safety thresholds and mechanisms of various types of tissue/cell/DNA damage, and to develop safe and effective radiation countermeasures for extended space travel. This proposal is to implement an already established multi-tissue platform to study the effects and mechanisms of space radiation and develop effective countermeasures for long missions. Over the last 8 years, we have bioengineered multiple human tissues starting from induced pluripotent stem cells (iPSCs) (heart, liver, bone, bone marrow, skin, sensory neurons, motor neurons, skeletal muscle, and midbrain). These tissues are matured and physiologically connected into an "organs on a chip" platform by vascular perfusion containing immune cells. With the addition of strong expertise in radiation biology, we collected preliminary data for the effects of photon and neutron radiation on sensory neurons, heart muscle, vascular endothelium, and bone marrow. Notably, the use of iPSCs allows individualized studies (e.g., for a specific astronaut). We now propose a radiation research platform consisting of four tissues: bone marrow (acute damage target), heart muscle (delayed damage target), liver (depo of granulocyte colony-stimulating factor (G-CSF)), and vascular perfusion with circulating cells. We further propose to evaluate an advanced nanoparticle-based modality for sustained delivery of G-CSF (a hematopoiesis stimulating factor) with oral delivery, or transactivation of the endogenous G-CSF gene for prolonged protection. These countermeasures will be tested against acute and fractionated high-linear energy transfer (LET) neutrons, simulated galactic cosmic rays, and photons (controls). We will validate the platform using iPSCs from healthy males and females and benchmark the collected data against known whole organism outcomes. The project will be milestone-driven and is expected to deliver a radically new approach enabling studies of space radiation damage and countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The impact of the studies conducted this far fall into three general areas, as itemized below. Radiation dosing: During deep space missions, astronauts will be exposed to a protracted mix of high-linear energy transfer (LET) (densely ionizing) and low-LET (sparsely-ionizing) radiations. While the low-LET radiations dominate by dose, it is anticipated that the high-LET radiations will dominate most radiation risks. This work therefore focuses on high-LET radiation exposures, with low-LET photon exposures as a reference radiation. The mission-relevant high-LET radiations are a mix of galactic cosmic rays (GCR) heavy ions as well as a significant component of fast neutrons, and our studies will use acute and fractionated fast neutrons generated at the Columbia University Radiological Research Accelerator Facility (RARAF) as a convenient surrogate for the overall high-LET component of shielded GCR. The rationale for using fast neutrons in this context is that the neutron energy distribution and high-LET distributions from the high-LET GCR component in a spacecraft in deep space are similar to those produced by the neutron facilities at our Columbia RARAF accelerator facility. In Year 3, as we will conduct multiple campaigns at Brookhaven National Labs, and continue to assess our "comparison" doses at Columbia RARAF and to use our photon exposures at Columbia's Radiological Research Center. Drug delivery and development: CS-PEI(PBA)-based nanocarriers is that they will provide a versatile platform for oral delivery of biologics with excellent efficiency, reduced toxicity, and prolonged protection of cells and tissues from irradiation, as compared to the commonly used polyethyleneimine and chitosan-based nanocarriers. The CS-PEI(PBA) polymer was shown to be tunable by modulating the grafting ratio of PEI and PBA group to achieve optimal efficacy and minimized cytotoxicity. Another impact of the tailored CS-PEI(PBA) nanocarriers is their potential for scavenging radiation-generated DAMPs. Radiation stress may induce DNA damage response and DNA repair, apoptosis, or immune responses. Thus, the incorporation of PBA and PEI moieties in the drug carrier design presents an intriguing strategy for controlling drug release in response to specific amounts of reactive oxygen species (ROS) and scavenging the DAMPs to suppress the cytokine storms. By virtue of its large size, pegfilgrastim exhibits size-based limitations in the penetration of biological barriers. The CS-PEI(PBA) polymer may actively improve the transcellular absorption of the pegfilgrastim. Further, this system may utilize oral delivery of PF for patient compliance and convenience. "Astronaut-on-a-chip": As we are conducting studies on assessing the effects of radiation on engineered tissues and establishing the full astronaut-on-a-chip system, we are showing the immense promise of in vitro human models for answering difficult questions in space health and biology. Although our models can only recapitulate certain aspects of radiation biology, we are hoping our efforts identify novel biomarkers and targets for the development of new therapeutics to protect astronauts in a potential Mars mission. In our described studies, we are targeting organ functions in those most sensitive to radiation damage (bone marrow), those with anticipated chronic changes (heart muscle), those that are a site for metabolism (liver), and those that may recapitulate continuous remodeling and tissue development (vasculature). Further, our platform and techniques are uniquely suited to answer new questions that arise in future space travel work, and are already overlapping with questions that have risen in related fields, such as cancer. We hope to continue to improve our organ-on-a-chip systems to better recapitulate biology, as well as better recapitulate the patient-specific nature of human health research.
	In this project, we developed a high-LET radiation research platform consisting of four tissues: bone marrow (acute damage target), heart muscle (delayed damage target), liver (site of metabolism), and vascular perfusion with circulating cells. Using this platform, we are evaluating the effects of simulated space radiation on human tissues and developing an advanced radiation countermeasure based on oral delivery of pegylated G-CSF (hematopoiesis stimulating factor), using sustained release nanoparticles. The radiation protection is being tested against mission-relevant doses of acute and fractionated high-LET fast neutrons, and acute and fractionated simulated galactic cosmic rays. AIM 1: ESTABLISH A HUMAN TISSUE MODEL TO STUDY MISSION-RELEVANT SPACE RADIATION. Our human tissue platform consists of bone marrow, heart muscle, and liver that are connected by vascular perfusion containing circulating immune cells. All tissues will be derived from the same iPS cells for individualized approaches to study the effects of space radiation and innovative countermeasures during long-term flights (e.g. Mars mission). Platform integration is enabled by the establishment of the optimal environment for each tissue, and separation of the tissue and vascular compartments by a selectively permeable endothelium, as in the body. Cell damage in each tissue, including the hematopoietic system, was evaluated over a period of 14 days, both for the individual tissues and for the integrated platform. In Aim 1, investigated the effects of low-LET photon radiation (acute exposure) and high-LET fast neutrons (acute and protracted exposure), both with a two-week follow-up. Our goal was to define mission-relevant dose and fractionation protocols for the month-long ground studies predictive of whole-body responses.

Task Progress:	AIM 2: TEST ADVANCED COUNTERMEASURES AGAINST ACUTE AND PROTRACTED HIGH-LET RADIATION. Protective countermeasures were based on pegylated granulocyte stimulating factor (pegfilgrastim, PF), an FDA-approved modulator of hematopoiesis. PF was tested in isolation, and in the platform's vascular flow in a naked form (to establish the dose and pharmacokinetics required for effective protection) and via controlled release nanoparticles (to achieve similar protection via oral delivery). The nanoparticles was designed to provide sustained release of PF in systemic circulation, with bone marrow and liver as target tissues. Both approaches are motivated by the need for oral (i.e. not weekly injected) countermeasure delivery. AIM 3: VALIDATE ADVANCED COUNTERMEASURES BY TESTING WITH SIMULATED GALACTIC COSMIC RAYS (GCRsim). Studies in Aim 1 and Aim 2, done in parallel, resulted in the detailed assessment of the effects of radiation on highly susceptible human tissues, and the effectiveness of the proposed radiation protection measures. In Aim 3, we demonstrated feasibility of conducting these studies in an individualized manner by pursuing the "astronaut on a chip" approach enabled through stem cell and tissue engineering technologies. Our goal here was to measure radiation damage under simulated galactic space radiation (acute and protracted) at the NASA Space Radiation Laboratory (NSRL). We demonstrated the utility of engineered human tissue models for studying the effects of high energy radiation, designed to mimic those seen in long-range space travel to Mars. We believe that our study is the first to use engineered human tissue models in a multi-OoC context for studying simulated galactic cosmic and neutron radiation exposures, and to establish a proof-of-concept platform and framework for using engineered human tissue models for mitigating other NASA "Red Risks" on Earth. Personalizing these model systems would allow for unprecedented understanding of how an astronaut's individual organ health may be impacted by space travel. It is critical, however, to compare data obtained in humanized radiation studies on Earth with those of past accidental nuclear exposures and animal studies. Human data have emerged from short-term studies of human cells in LEO experiments on the ISS, as well as data collected from astronauts after returning to Earth.

Bibliography Type:	Description: (Last Updated: 07/15/2026)
Articles in Peer-reviewed Journals	Tavakol DN, Nash TR, Kim Y, He S, Fleischer S, Graney PL, Brown JA, Liberman M, Tamargo M, Harken A, Ferrando AA, Amundson S, Garty G, Azizi E, Leong KW, Brenner DJ, Vunjak-Novakovic G. "Modeling and countering the effects of cosmic radiation using bioengineered human tissues." Biomaterials. 2023 Oct;301:122267. https://doi.org/10.1016/j.biomaterials.2023.122267 ; PMID: 37633022; PMCID: PMC10528250 , Oct-2023
Articles in Peer-reviewed Journals	Ronaldson-Bouchard K, Baldassarri I, Naveed Tavakol D, Graney PL, Samaritano M, Cimetta E, Vunjak-Novakovic G. "Engineering complexity in human tissue models of cancer." Adv Drug Deliv Rev. 2022 May;184:114181. Review. https://doi.org/10.1016/j.addr.2022.114181 ; PMID: 35278521; PMCID: PMC9035134 , May-2022
Articles in Peer-reviewed Journals	Tavakol DN, Chen J, Chavkin NW, Tavakol TN, Hirschi KK, Vunjak-Novakovic G. "Lessons from Biology: Engineering design considerations for modeling human hematopoiesis." Current Stem Cell Reports. 2021 Dec;7(4):174-84. https://doi.org/10.1007/s40778-021-00195-5 , Dec-2021
Articles in Peer-reviewed Journals	Tavakol DN, Nash TR, Kim Y, Graney PL, Liberman M, Fleischer S, Lock RI, O'Donnell A, Andrews L, Ning D, Yeager K, Harken A, Deoli N, Amundson SA, Garty G, Leong KW, Brenner DJ, Vunjak-Novakovic G. "Modeling the effects of protracted cosmic radiation in a human organ-on-chip platform." Adv Sci (Weinh). 2024 Jul 4:e2401415. https://doi.org/10.1002/advs.202401415 ; PMID: 38965824 , Jul-2024
Articles in Peer-reviewed Journals	He S, Zhu Y, Tavakol DN, Ye H, Lao YH, Zhu Z, Xu C, Chauhan S, Garty G, Tomer R, Vunjak-Novakovic G, Zou J, Azizi E, Leong KW. "Squidiff: Predicting cellular development and responses to perturbations using a diffusion model." Nat Methods. 2026 Jan;23(1):65-77. https://doi.org/10.1038/s41592-025-02877-y ; PMID: 41184550; PMCID: PMC12872408 , Jan-2026
Awards	Vunjak-Novakovic G. "National Academy of Engineering, Chair of Section 2: Bioengineering. " Jan-2023
Awards	Vunjak-Novakovic G. "New York's Top 50 in Biotechnology Honoree. " Jun-2021
Awards	Vunjak-Novakovic G. "Order of Karadjordje Star, Second Class. " Apr-2021
Awards	Vunjak-Novakovic G. "Rosalind Franklin Society and Mary Ann Liebert Inc., Annual Award for the Best Paper in Stem Cells and Development. " Feb-2023
Awards	Tavakol DN. "“Audience Choice” Award, Columbia 3 Minute Thesis Competition. " Feb-2023
Awards	Tavakol DN. "Elsevier, Cell Press Underrepresented Minority Travel Grant. " Dec-2022
Awards	Tavakol DN. "International Society for Experimental Hematology (ISEH) Ihor Lemishka Travel Grant. " Sep-2022
Awards	Tavakol DN. "International Society for Experimental Hematology (ISEH) T. Ray Bradley Award; 2nd place oral talk at annual meeting. " Aug-2023
Awards	Tavakol DN. "NIH Ruth L. Kirschstein Individual Predoctoral Fellowship (F31). " Jul-2022
Awards	Tavakol DN. "Rising Star in Engineering in Health, 1 of 20 early career scientists chosen by Johns Hopkins and Columbia Universities. " Nov-2022
Awards	Tavakol DN. "Tissue Engineering & Regenerative Medicine International Society Americas Travel Grant. " Jul-2022
Awards	Tavakol DN. "Yuen-huo Hung and Chao-chin Huang Award in Biomedical Engineering; Outstanding Graduating Doctoral Student (2023). " May-2023

Awards	Vunjak-Novakovic G. "AIMBE Pierre Galletti Award. " Mar-2021
Awards	Vunjak-Novakovic G. "Columbia Tech Ventures Ambassador " Jun-2023
Awards	Vunjak-Novakovic G. "Elected to the Royal Society of Canada – Academy of Science." Jun-2023
Awards	Vunjak-Novakovic G. "European Patent Office, Inventor Prize Finalist and Popular Prize Award Winner. " Jun-2021
Awards	Vunjak-Novakovic G. "International Academy of Medical and Biological Engineering Fellow. " May-2021
Awards	Vunjak-Novakovic G. "Lifetime Achievement Award, Tissue Engineering and Regenerative Medicine Society (TERMIS). " May-2021
Awards	Leong K. "Member, National Academy of Medicine. " Jun-2021